Capstone Project – I: Review – 2 on

Voice-Based Chatbots

Presented by the Batch:

Y Sathwik – VU22CSEN0101112
B Sai Vardhan – VU22CSEN0102342
N Samuel Ricky – VU22CSEN0101727
G Srivatsa – VU22CSEN0101391

Under the Guidance of

Muddada Murali Krishna

Assistant Professor

•

Department of Computer Science and Engineering
GITAM School of CSE, GITAM (Deemed to be University),
Visakhapatnam.

CONTENTS

1. **Recap of Capstone Project Introduction Review-1**
   A. Problem Statement
   B. Objectives
   C. Feedback from Guide/Reviewers and actions taken after Review–1
2. Requirement Analysis
3. Identification of tools/technology/algorithms
4. Design Strategy (project workflow/ system design/ DFD/UML design)
5. Conclusion
6. References

Problem Statement

<u>Problem:</u> Despite robust digital infrastructure, campus data remains fragmented and hard to access for many users. Students, faculty, and parents often struggle with:

- Complex portals and apps
- Slow/manual data search
- Lack of personalized responses
- Limited hands-free or non-digital access

<u>Goal:</u> Develop a unified, voice-driven AI interface to college data, automating query responses for every campus stakeholder, with robust security and full role-based data separation.

Objectives

What do we wish to achieve?

1. Create a real-time, voice-based query assistant for all campus users.

2. Integrate deeply with the college database, respecting all user roles and permissions.

3. Provide universal access: via web platform, mobile app, and telephone call-to-query.

4. Ensure future extensibility for agentic workflows (support, advisory, scheduling, notifications).

Feedback from Guide/Reviewers and actions taken after Review–1

Feedback from Guide/Reviewers

1. Suggested to find out an efficient way to retrieve required data & answer questions.
2. Design a Modular Architecture instead of monolithic.

Actions taken

1. Used n8n tool to solve this issue, which connects with the database and uses AI to retrieve the relevant data.
2. Architectured a Modular High-Level Design (HLD) & DFD.

Requirement Analysis

Hardware Requirements

- **Server/Cloud Infrastructure:** AWS, Digital Ocean or Azure VMs | Minimum: 2GB RAM, 2 vCPU for testing; 4GB+ for production deployment
- **Database Server:** MongoDB Atlas (cloud-hosted, free tier sufficient for 1M documents)
- **n8n Workflow Engine:** Self-hosted on Docker container (1GB RAM minimum)
- **Audio Processing Server:** Whisper/Coqui TTS GPU acceleration optional (CPU adequate for pilot)
- **Client Devices:** Standard laptops, tablets, smartphones with modern browsers (Chrome, Firefox, Safari, Edge)
- **Microphone/Audio Input:** Built-in or USB microphone for voice capture

Requirement Analysis

Software Requirements

- **Backend Framework:** FastAPI (Python 3.9+) for API routing and session management
- **Frontend Framework:** Next.js (React 18+) with TypeScript for PWA and responsive UI
- **Workflow Orchestration:** n8n (self-hosted via Docker) for AI agent management and integrations
- **Speech-to-Text:** OpenAI Whisper (open-source model, self-hosted or API)
- **Text-to-Speech:** Coqui TTS (open-source, self-hosted) for natural voice synthesis
- **LLM/AI Model:** Llama 2/3 (via Ollama) or Gemini API for query understanding and response generation
- **Vector Database:** ChromaDB or Weaviate (self-hosted, free open-source) for knowledge retrieval
- **Primary Database:** MongoDB Atlas (free tier: 512MB storage, scalable to production)
- **Version Control:** Git/GitHub for codebase management
- **Containerization:** Docker and Docker Compose for deployment consistency
- **API Gateway:** n8n webhooks + FastAPI routes for request handling
- **Operating Systems:** Linux , macOS, or Windows with WSL2

Identification of tools/technology/algorithms

Tools

Workflow & Orchestration:

• **n8n**: Visual workflow builder for AI agent management, multi-source data integration, and conversational logic

• **MongoDB Atlas:** Cloud database for structured student/faculty/course data with flexible schema

Backend Development:

• **FastAPI**: Asynchronous REST API framework for session routing, user management, and webhook integration

• **Docker:** Containerization for consistent deployment across environments

• **Web Audio API:** Browser-native speech capture and playback for voice interface

Speech Processing:

• **OpenAI Whisper**: Open-source STT model with 99%+ accuracy across accents and languages

• **Coqui TTS**: Lightweight, open-source text-to-speech synthesis with emotional modulation

AI/ML & LLM:

• **Gemini API (or Llama via Ollama):** Large language model for natural language understanding and response generation

• **Vector Store:** (ChromaDB/Weaviate): Semantic search and knowledge base retrieval

Deployment & Hosting:

• **AWS/Digital Ocean / Azure:** Cloud infrastructure using available student credits

• **GitHub:** Version control and CI/CD pipeline management

Identification of tools/technology/algorithms

Technologies

Real-Time Communication:

- **WebSocket:** Bidirectional audio streaming and instant response delivery
- **Webhooks:** n8n to FastAPI communication for query resolution and context updates

Data Processing & NLP:

- **Transformer-Based Models:** Whisper for speech recognition, Llama/Gemini for semantic understanding
- **Vector Embeddings:** Dense representations for efficient knowledge retrieval and similarity matching
- **Context Management:** Conversation history stored in MongoDB for multi-turn dialogue support

Security & Authentication:

- **Role-Based Access Control (RBAC):** MongoDB schemas enforce student/faculty/admin data separation
- **JWT Tokens:** Stateless session management for web and mobile
- **TLS/SSL Encryption:** End-to-end encryption for all API calls and audio streams

Scalability:

- **Horizontal Scaling:** Docker swarms or Kubernetes for load distribution
- **Caching:** Redis for session and frequently accessed data (e.g., faculty lists, schedules)
- **Database Indexing:** MongoDB indices on roll_number, course_code for sub-100ms queries

Identification of tools/technology/algorithms

Algorithms

Information Retrieval:

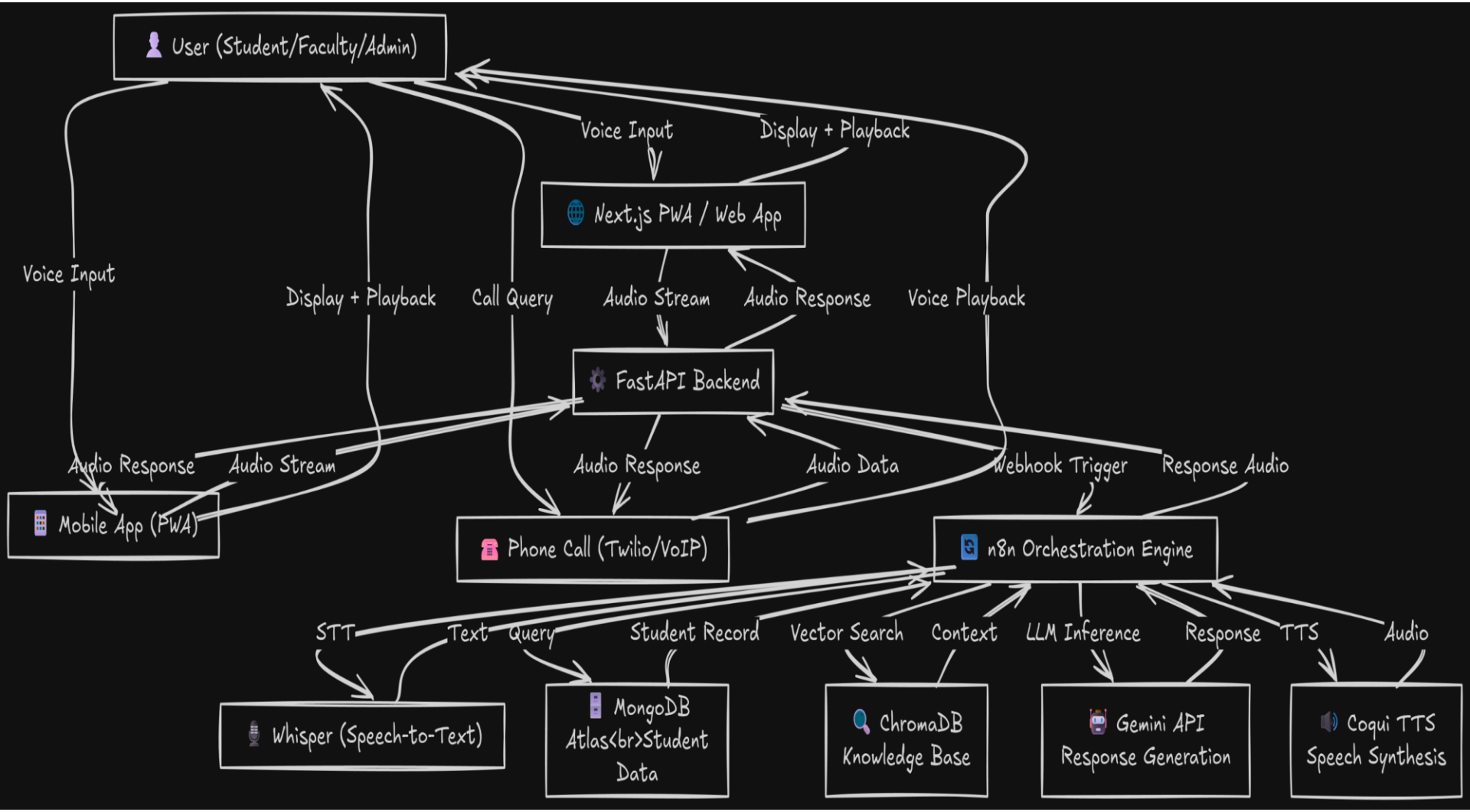
- **Vector Search:** ChromaDB/Weaviate for fast semantic matching
- **Hybrid Retrieval:** Combine keyword search (MongoDB) with semantic search (vector DB)
- **Ranking Algorithms:** Re-ranking algorithms to surface most relevant results.

Dialogue Management:

- **State Machine:** n8n workflows for managing multi-turn conversation flow
- **Slot Filling:** Extract and validate required parameters (e.g., semester, course code) before executing query
- **Context Window Management:** Sliding window of last N messages from the user.

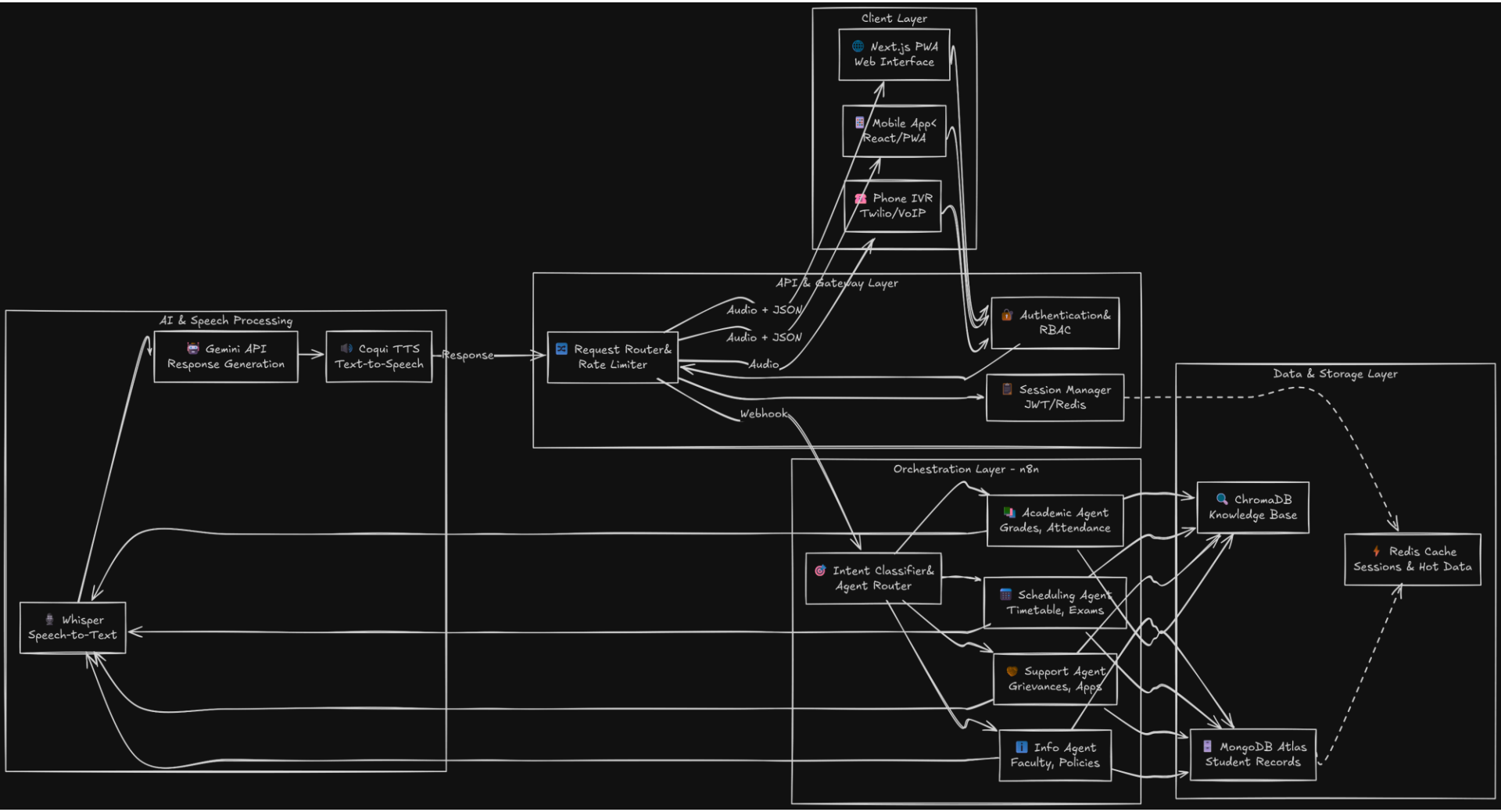
Design Strategy

Project Workflow



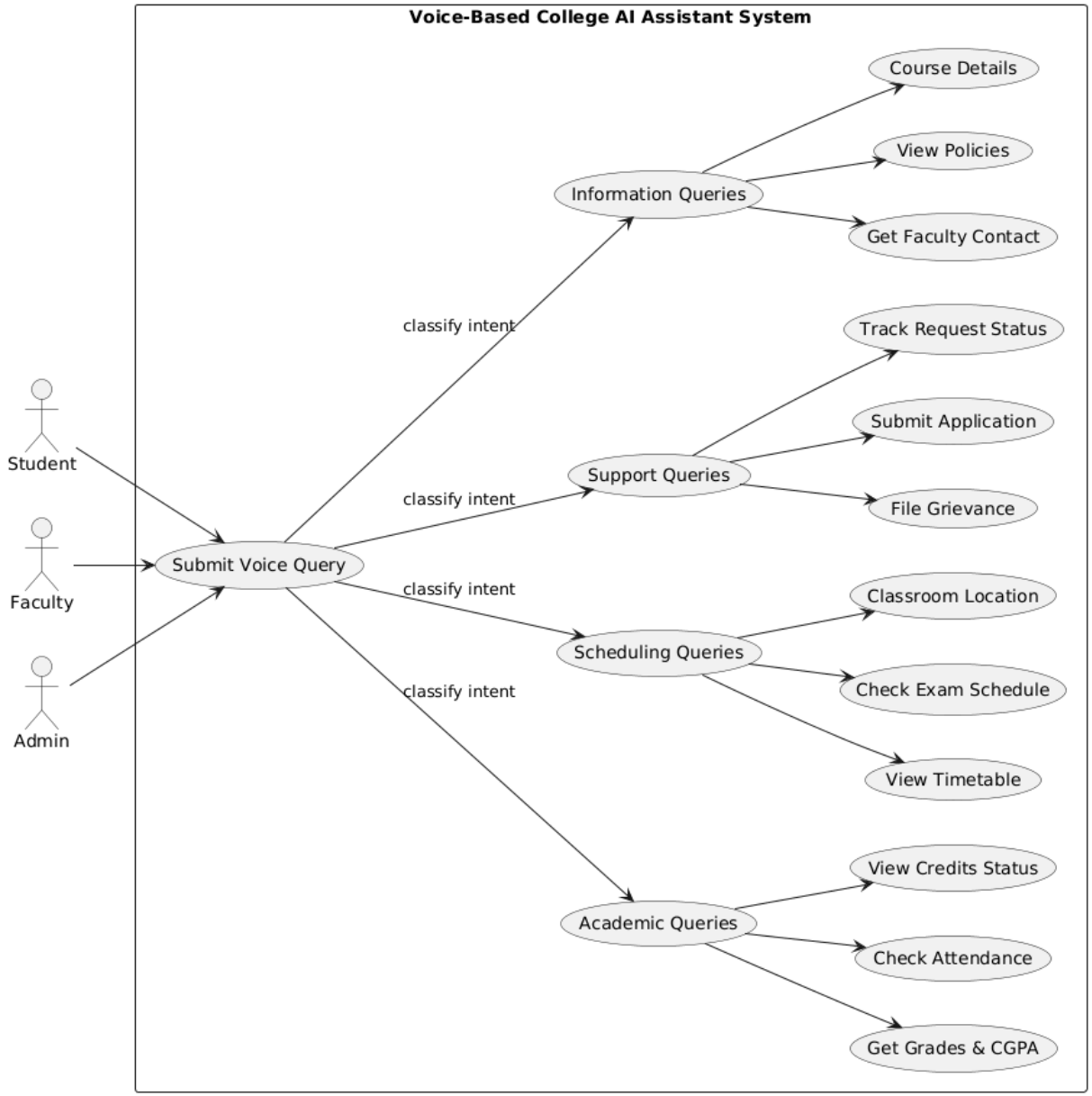
Design Strategy

System Design – HLD



Design Strategy

Use Case Diagram



Conclusion

GITAM
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Summary of Progress

- **Architecture Finalized:** Modular, scalable design with n8n orchestration at the core; FastAPI backend; Next.js frontend
- **Database Designed & Deployed:** Comprehensive MongoDB schema created and uploaded to Atlas; tested with sample student record.
- **Technology Stack Locked:** All free/open-source tools selected (Whisper, Coqui TTS, Llama/Gemini, ChromaDB, n8n)
- **Workflow Orchestration Planned:** n8n agents mapped for academic, scheduling, support, and info domains
- **Security Framework:** Role-based access control (RBAC), JWT authentication, TLS encryption implemented
- **Deployment Infrastructure:** Hosting on AWS/Digital Ocean/Azure

Team's Confidence on Project Completion

- Very High Confidence - Team has:
- Comprehensive technical knowledge of n8n, FastAPI, Next.js, and MongoDB
- Clear separation of work across 4 team members (architecture, DB, frontend, QA)
- Risk mitigation strategies (fallback to text if TTS fails; cached responses for high-load queries)
- Realistic timeline: 12–16 weeks for MVP (core academic queries + web interface)

Conclusion

GITAM
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Expected Outcomes & Impact

Immediate Impact (if launched):

- **Students:** Instant, hands-free access to grades, attendance, schedules, faculty info via voice
- **Faculty:** Easy query of student rosters, performance analytics, event management via voice
- **Administration:** Streamlined support ticket filing, policy clarifications, all via conversational AI

Scalability & Future Roadmap:

- **Expand to multi-campus deployment** (replicate MongoDB + n8n workflows per campus)
- **Add real-time integration** with campus management systems (hostel, sports, events)
- **Introduce emotional intelligence:** detect stress/frustration in student voice, suggest counseling resources
- **Multi-language support:** auto-translate queries and responses to regional languages
- **Advanced analytics:** insights on most-asked questions, academic bottlenecks, student satisfaction trends

References

1. n8n Official Documentation

https://docs.n8n.io/

Comprehensive resource for understanding n8n workflow automation, API integrations, and agent mode orchestration used as the AI engine backbone.

2. OpenAI Whisper Speech-to-Text Model

https://openai.com/research/whisper

Open-source, powerful speech recognition model utilized for converting voice queries to text accurately across multiple languages and accents.

3. Coqui TTS - Open Source Text-to-Speech

https://github.com/coqui-ai/TTS

Open source, scalable text-to-speech synthesis used to convert AI-generated text responses back to natural voice output.

4. FastAPI Documentation

https://fastapi.tiangolo.com/

Async Python web framework for high-performance REST APIs

5. Next.js PWA Guide

https://nextjs.org/learn/pwa/introduction-to-pwa

Building progressive web apps with installability, offline support, and real-time features

6. Gemini API Documentation

https://ai.google.dev/docs

LLM inference, prompt engineering, and integration guides